

below average outliers. This will result in actual costs being higher than the predicted cost. For example, if we trade 200,000 shares out of a total ADV of 1,000,000 shares, we may be tempted to state that a full day strategy corresponds to a trading rate of 20%. However, if the actual volume on the day is only 800,000 shares, the actual trading rate will be 25%, resulting in higher than predicted costs. The market impact forecasting error will be biased (to the high side) when using the ADV measure to predict daily volume. In our sample, we found the ADV to be higher than the actual volume on the day 65% of the time.

Step 2. Estimate the *Day Of Week(t)* parameter.

We analyzed whether or not there is a cyclical trading pattern during the week. To avoid bias that may be caused by special event days such as FOMC, triple witching, index reconstitution, earnings, month end, etc., we adjusted for these days in our analysis. It is important to note that if month end is not excluded from the data there may be a strong bias suggesting that Friday is the heaviest trading day of the week, since three out of seven month ends occur on Fridays (due to weekends). Many investors trade more often on the last day of the month.

Our day of week process is as follows:

- For each stock compute the percentage of actual volume traded on the day compared to the average volume in the week.
- Exclude the special event days that are historically associated with higher traded volume.
- Compute the average percentage traded on each day across all stocks in the sample.
- It is important to use a large enough sample in the analysis. We used one full year trading period to compute the day of week effect.

The result of our day of week analysis is shown in [Figure 7.4](#). Monday is consistently the lowest volume day in the week for large and small cap stocks. After adjusting for month end volume, we found that small cap volume increases on Fridays but large cap volume decreases. The effect may be due to investors not being willing to hold an open position in small cap stocks over the weekend for fear there is too much market exposure for small cap stocks and therefore they may elect to pay higher market impact before weekend to ensure completion.

Conclusion 2

- Stock trading patterns exhibit a cyclical weekly pattern.
- The cyclical pattern is different for large cap and small cap stocks.